AP PRECALCULUS

Unit 1 • Worksheet 1.3B • Answer key

Worksheet 1.3B • Rates of Change in Linear and Quadratic Functions

Answer key with worked solutions

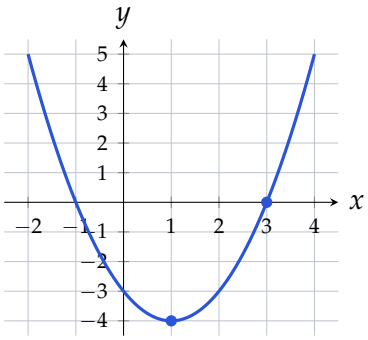
Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

The graph of a quadratic f is shown. What is the average rate of change of f on $[1, 3]$?



(A) -2

(B) 4

(C) 2

(D) 0

Solution

Answer: (C). $f(1) = -4$, $f(3) = 0$: $\frac{4}{2} = 2$.

Question 2

NO CALCULATOR

For the same function, the average rate of change on $[-1, 1]$ is -2 and on $[1, 3]$ is 2. Which statement explains this?

(A) f is linear.

(B) The rate of change of f is increasing, so the graph is concave up.

(C) The rate of change of f is decreasing.

(D) f has a maximum at $x = 1$.

Solution

Answer: (B). Average rates over consecutive equal intervals increase: concave up. The vertex $(1, -4)$ is a minimum.

Question 3

NO CALCULATOR

At what value of x does the rate of change of $f(x) = 3x^2 - 12x + 1$ change from negative to positive?

(A) $x = 2$
(B) $x = -2$
(C) $x = 4$
(D) $x = 12$

Solution
Answer: (A). Vertex at $x = \frac{12}{6} = 2$, and $a > 0$.

Question 4 (Free response)

NO CALCULATOR

The height of a thrown ball is recorded each second: $h(0) = 2$, $h(1) = 17$, $h(2) = 22$, $h(3) = 17$, $h(4) = 2$ (meters).

(a) Find the average rates of change on $[0,1]$, $[1,2]$, $[2,3]$, $[3,4]$. What do you notice?
(b) Write a quadratic model $h(t)$.
(c) Is the graph of h concave up or concave down? Interpret the rate of change in context.

Solution
(a) 15, 5, -5, -15: they decrease by 10 each second.
(b) $2a = -10$, $a = -5$; vertex $(2, 22)$: $h(t) = -5(t-2)^2 + 22$. Check $h(0) = -20 + 22 = 2$.
(c) Concave down. The ball's upward velocity decreases by about 10 m/s each second; after $t = 2$ the ball falls faster and faster.